

Replicating the paper: *exact title of the paper you are replicating, followed by the citation, e.g. Bitetto et al. (2024)*

*Your name**

September 21, 2026

Abstract

Briefly explain what the paper you are replicating does, what it contributes, and how it bridges the gap left by the papers before it. Then say why you are interested in this paper. Write it as one paragraph of continuous prose, not as a list. Aim for 150–200 words. Do not simply copy the original abstract: the abstract of a replication describes the replication.

Keywords: *copy the keywords from the paper you are replicating*

JEL Codes: *copy the JEL codes from the paper you are replicating. If the paper does not give any, choose from the JEL classification system and say in a footnote that you chose them yourself.*

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1 Motivation

Treat this as a short journal article rather than a homework write-up. The whole manuscript is limited to 2,400 words, excluding tables, figures and references, so every paragraph has to earn its place. This file already uses the SSRN format the lab writes in, so if you later develop the work into a thesis or a conference paper you will not have to move it.

Your introduction should do four things, in this order. Say what question the original paper asks and why it matters. Say what the paper found. Say what you set out to reproduce, and be explicit about what you could and could not obtain — the same data, a subset, or a substitute. Finally, say what a reader learns from your replication that they would not learn from the paper alone.

Cite five papers that are fundamental to the question, and five that attack it with machine learning or AI. Follow the Google Scholar convention for bibkeys, as in `bitetto2024can`. A citation you have not read does not belong here.

2 Data

2.1 Dependent and independent variables

List every variable with its acronym, a one-line description, and its type: binary, categorical, or continuous. State clearly which variable is the response and which are the predictors. If your data differs from the data in the original paper, say so here and say what you expect that to change.

2.2 Data exploration, cleaning and preprocessing

2.2.1 Cross-sectional data

- Check whether there are missing values.
 - It is not sufficient to list the proportion of missing values per variable.
 - Show exactly which observations and which variables the missing values fall in.
 - If values are missing, consider a missing indicator.
- Check whether there are outliers.
 - Check first whether they are errors introduced when the data was retrieved. If they are not, keep them for the analysis.

- For categorical variables, show the frequency tables.
 - For predictors, decide whether to use dummy variables or to treat the variable as continuous, and whether to merge categories that are extremely rare.
 - For the response, decide between logistic and multinomial regression, and check whether the classes are imbalanced.
- For continuous variables, report mean, standard deviation, minimum, maximum, skewness and kurtosis.
 - Decide whether a logarithm is needed, which it usually is when the data is heavily skewed.
- Scaling matters. Min-max or z-score puts the features on a common range and makes the numerical procedures more stable. Report whether you scaled or not.
- Use pairwise scatter plots to look for patterns, and decide whether quadratic terms, two-way interactions or logarithms are called for.
- Show a heatmap of the correlations.

3 Methodology

3.1 Benchmark model

Use the same benchmark the original paper used, so that your numbers are comparable to theirs. If you cannot, say which model you substituted and why.

One of the following:

- Logistic regression
- Multinomial regression
- Linear regression

3.2 Experimental Design

Give a flow chart of your procedure, from raw data to reported number. A reader should be able to reproduce your result from that figure alone.

- Explain the design of the replication.

- State what is novel in your version. For a replication this is usually not a new method: it is a test the original paper did not run, data it did not use, or an assumption it did not check.

3.3 Performance measures

Use the measures the original paper reports, so the comparison is direct. Add any measure you think the paper should have reported, and say why.

4 Results

Put your numbers and the original paper's numbers side by side in one table. Then state plainly whether the result replicated. "It did not replicate" is a finding, not a failure, provided you can show what you did.

5 Conclusion

Two paragraphs. What survived the replication, and what a reader should now believe less strongly than before.

Declaration

Declaration of Pedagogical Purpose

This manuscript is prepared as part of the course project for Machine Learning and FinTech at National Yang Ming Chiao Tung University, instructed by Professor Huei-Wen Teng. The work is intended solely for pedagogical purposes, including learning, discussion, and academic training. It does not represent original research submitted for journal publication, nor should it be considered professional financial advice.

Declaration of generative AI and AI-assisted technologies in the writing process

Name the tools you actually used. Replace the bracketed placeholders below.

During the preparation of this work, the authors used [ChatGPT] and [Grok], developed by [OpenAI] and [xAI] respectively, in order to improve the clarity, precision, and overall

quality of the writing. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the article.

References

Bitetto, A., P. Cerchiello, S. Filomeni, A. Tanda, and B. Tarantino (2024). Can we trust machine learning to predict the credit risk of small businesses? *Review of Quantitative Finance and Accounting* 63(3), 925–954.

A *Appendix title*

Titles have to be concise. Put here anything a reader needs in order to reproduce your work but does not need in order to follow the argument.